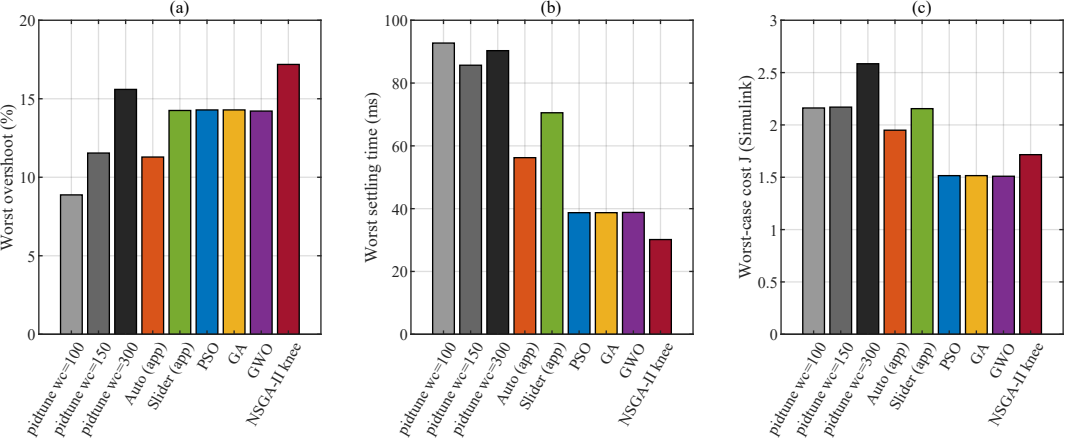

(a)
Worst overshoot (%)
0
5
10
15
20
(b)
Worst settling time (ms)
0
20
40
60
80
100
(c)
Worst-case cost J (Simulink)
0
0.5
1
1.5
2
2.5
3
pidtune wc=100
pidtune wc=150
pidtune wc=300
Auto (app)
Slider (app)
PSO
GA
GWO
NSGA-II knee